GEO-SPATIAL HAZARD ZONATION MAP OF FLOOD IN GOLAGHAT DISTRICT, ASSAM , INDIA USING GIS

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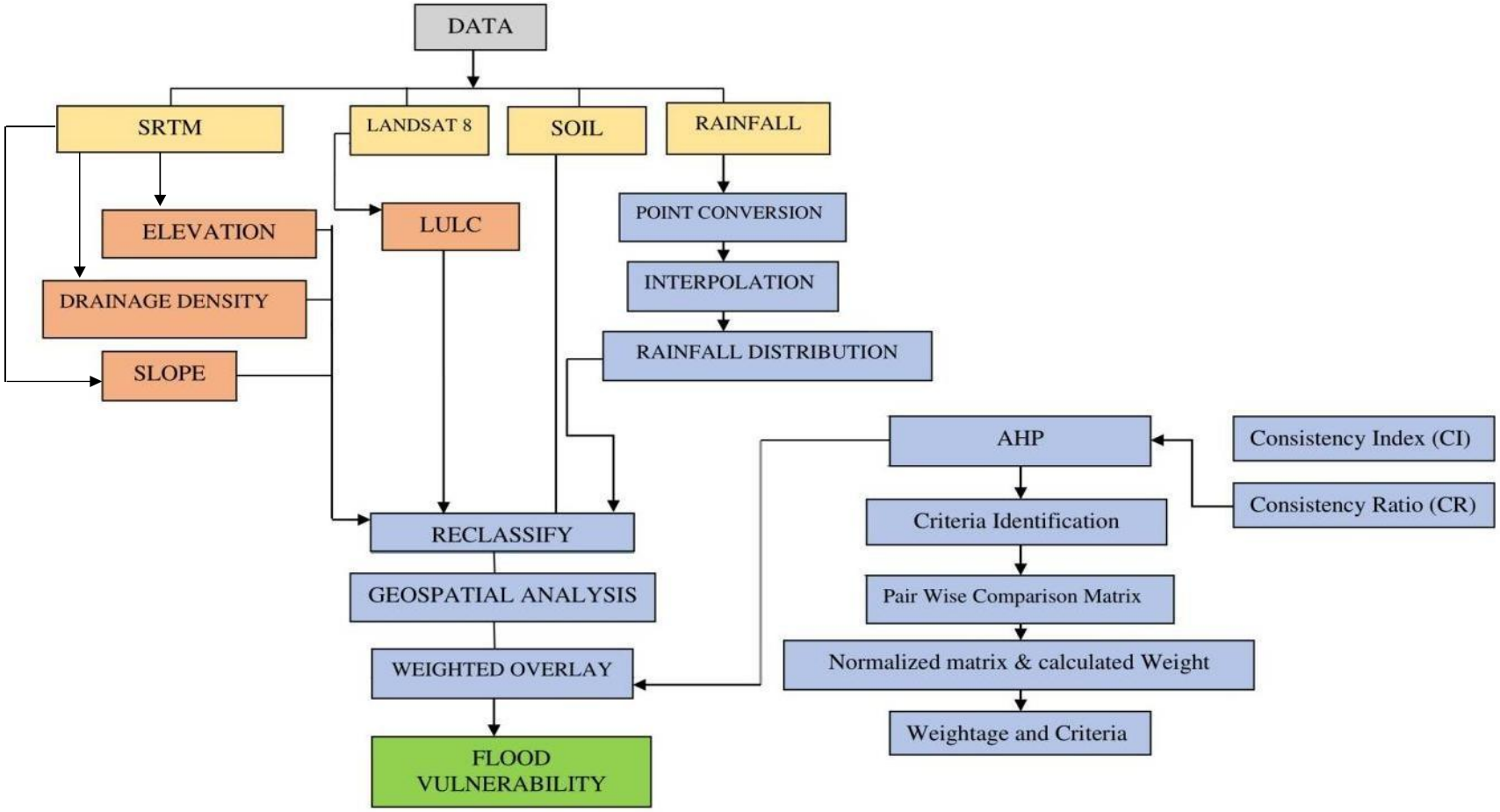
Aim

Based on Flood hazard zonation map to identify the flood vulnerable areas in Golaghat Districts in Assam

Methodology (Using AHP)



Sources of Data

List of NGOs and Organizations

- Assam Disaster Management Authority
- Indian Meteorological Department
- Flood Reporting and Information Management system
- SEED INDIA
- Akshay Patra Foundation
- Assam Flood Relief
- ActionAid.org

Data set	Resolution	Data source
LANDSAT-8 Satellite image	m	USGS Earth Explorer
SRTM DEM	m	USGS Earth Explorer
Rainfall	(0.25° x 0.25°)	IMD (India Meteorological Department)
Soil Texture data	1:50,000	NBSS & LUP (The National Bureau of Soil Survey and Land Use Planning)

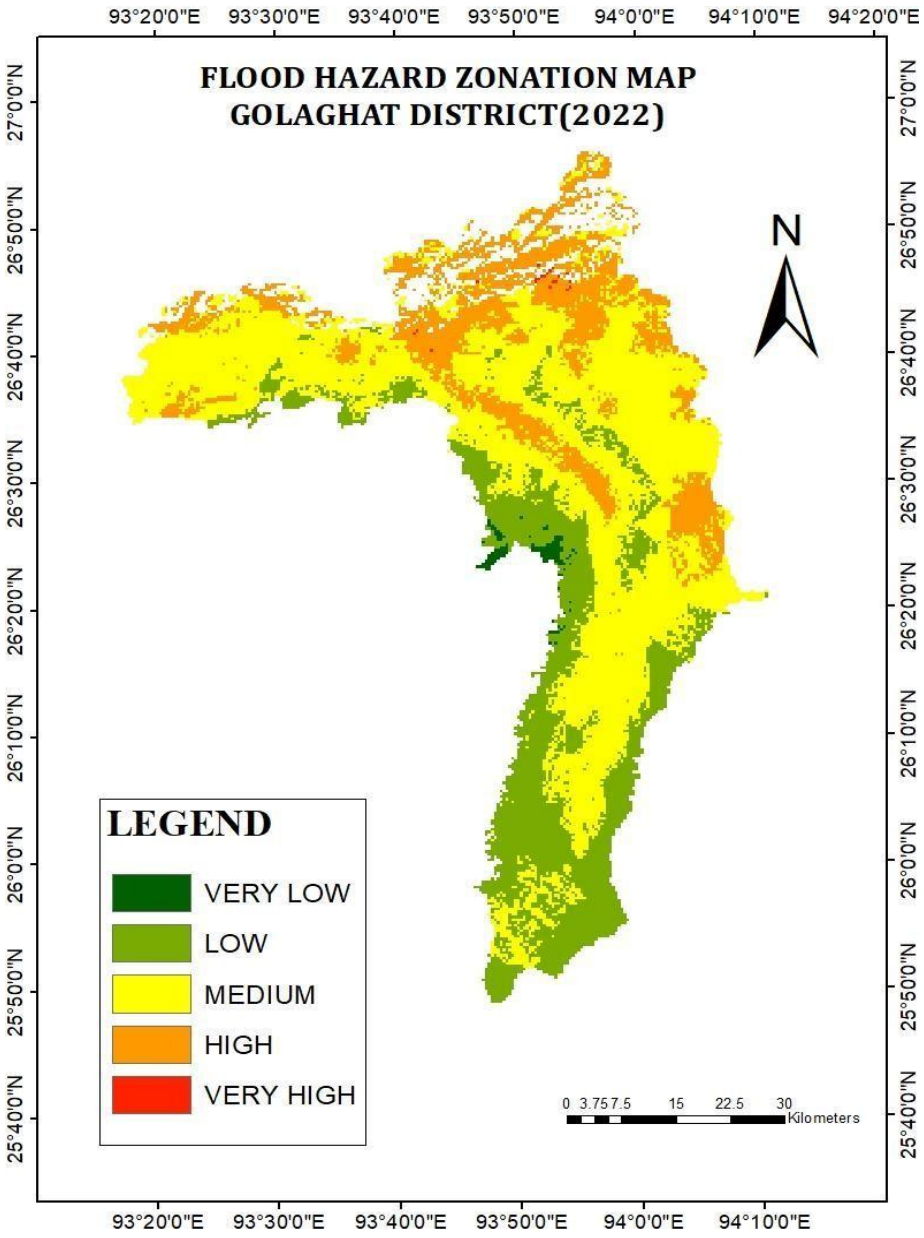
Flood Vulnerability Assessment

Pairwise comparison matrix for the selected physical features of Golaghat District

Parameters	Rainfall	LULC	DD	SLOPE	Elevation	SOIL
Rainfall	1.00	3.00	3.00	7.00	5.00	5.00
LULC	0.33	1.00	0.33	3.00	3.00	3.00
DD	0.33	3.00	1.00	7.00	3.00	4.00
SLOPE	0.14	0.33	0.14	1.00	0.50	0.50
Elevation	0.20	0.33	0.33	2.00	1.00	2.00
SOIL	0.20	0.33	0.25	2.00	0.50	1.00

CR = 0.04

FLOOD HAZARD ZONATION MAP


FLOOD HAZARD ZONATION MAP
GOLAGHAT DISTRICT(2022)
N
LEGEND
VERY LOW
LOW
MEDIUM
HIGH
VERY HIGH
0 3.75 7.5 15 22.5 30
Kilometers
93°20'0"E
93°30'0"E
93°40'0"E
93°50'0"E
94°0'0"E
94°10'0"E
94°20'0"E
27°0'0"N
26°50'0"N
26°40'0"N
26°30'0"N
26°20'0"N
26°10'0"N
26°0'0"N
25°50'0"N
25°40'0"N